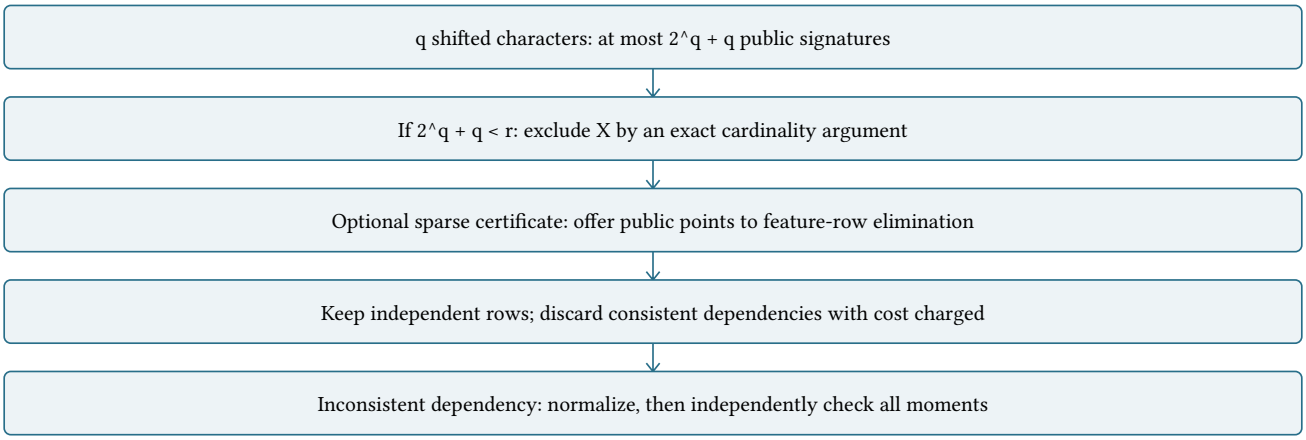
Character-signature exclusion and sparse selection

2026-11-03 · Derived theorem / source construction · Numerical trials: zero

Structural bound. Outside q exceptional character-zero points, the q shifted quadratic characters have binary signs. Thus K = min(r, 2^q + q) bounds the number of signatures. Any function of the signature agrees with X at most once per signature fiber, hence at most K times. If K < r, representation of X is excluded at every polynomial degree, independently of Gram weights.



Public selection. A bounded streaming routine evaluates the original cubic features at supplied points. It retains independent rows and their original-row combinations, applying the same elimination to target values. A zero feature residual with nonzero target produces a normalized sparse dual. A separate checker recomputes all moments and the pairing from public support.

Progress proof. At each nonterminal state, some feature combination fits the retained independent rows. At least r−K points disagree with that fit. Each such point either raises feature rank or produces a certificate. Under independent uniform point sampling and K < r, conditional progress probability is at least delta=(r−K)/r. Expected offers are at most (m+1)/delta, where m=1+q+C(q,2)+C(q,3). No independence of character signs is assumed.

Derived quantity	Bound / schedule
Witness support	At most m+1 public points
Elimination at rank t	t(m+2)+t(t−1)/2 products and subtractions
Stored field entries	tm+t+t(t+1)/2
Deterministic sufficient stream	K+1 distinct points, only if K < r
Sampler / stopping	Caller-supplied stream; explicit finite cap

Feature evaluation, normalization, inversions, native work and independent checking are additional and accounted in DERIVATION.md. The theorem precheck needs no field operations. Exhausted samples mean inconclusive. No automatic field enumeration, random sampler or signature collision table is introduced.

Applicability. The saved F479 width-15 failure has K=r, so the strict bound is inconclusive there. Its earlier dual certificate remains necessary. The retained recovery panels start outside this guaranteed regime; no saved-panel speed improvement is claimed. Positive represented programs remain separate.

Evidence / next. Syntax and preserved source bytes checked statically; no numerical, corruption, heap or performance trials. Next exploit cubic degree to sharpen agreement or certify an interpolant when K=r. A sampled fit alone cannot establish a representation valid on the full field.

Sources. NIST DLMF 27.9 (Legendre alphabet including zero); preserved feature-span-dual report and coefficient construction. Cardinality and progress arguments are derived here, without a novelty claim for general row elimination.